

ALEXANDRIA INTL, LA, USA

WMO: 747540

Lat: 31.335N

Lon: 92.559W

Elev: 84

StdP: 14.65

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 93915

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.9	29.8	14.2	11.3	33.8	18.7	14.1	37.8	23.7	53.7	20.3	57.2	4.8	10	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.9	97.2	76.8	95.0	76.8	93.0	76.6	80.3	89.4	79.5	89.2	78.7	88.3	6.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.4	147.7	83.7	77.1	141.6	83.2	76.2	137.4	82.5	43.7	89.3	43.0	89.2	42.2	88.5	91.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	16.9	14.9	21.1	100.5	3.8	3.3	18.4	102.9	16.1	104.8	14.0	106.6	11.3	109.0
			19.3	83.4	3.6	3.1	16.7	85.6	14.6	87.5	12.6	89.2	10.1	91.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.3	49.8	53.5	60.0	67.0	75.1	80.6	82.6	82.8	78.1	68.0	57.8	51.3
	DBStd	14.10	10.27	10.18	8.83	7.03	5.53	3.54	2.89	3.35	4.93	7.96	8.76	9.84
	HDD50	360	134	72	21	1	0	0	0	0	0	1	27	104
	HDD65	1837	481	342	201	60	5	0	0	0	1	63	245	439
	CDD50	6670	126	169	332	511	779	917	1011	1018	843	557	262	145
	CDD65	2673	8	19	48	120	319	467	546	553	394	155	30	14
	CDH74	26175	20	83	279	946	2928	4717	5815	5978	3738	1403	216	52
	CDH80	11596	0	5	30	200	1135	2209	2808	3024	1703	451	29	2
Wind	WSAvg	6.1	7.3	7.6	7.7	7.6	6.5	5.2	4.2	4.1	4.9	5.3	6.1	7.0
Precipitation	PrecAvg	59.8	5.6	4.9	5.5	4.6	5.0	4.9	4.4	4.1	4.1	4.5	5.7	6.5
	PrecMax	87.4	13.9	13.2	13.2	14.8	11.9	13.8	13.3	10.4	11.9	16.1	17.5	20.8
	PrecMin	36.1	1.0	0.5	0.5	0.3	0.0	1.0	0.4	0.1	0.4	0.0	0.2	1.7
	PrecStd	11.6	3.3	2.6	3.0	3.4	2.9	3.1	2.6	2.4	2.4	4.0	3.3	3.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.7	80.8	83.0	88.1	93.2	98.7	98.9	100.4	97.2	90.9	83.6	79.1
		MCWB	66.4	67.4	67.7	71.9	74.1	76.5	77.7	76.8	74.7	74.1	70.5	69.9
	2%	DB	73.1	76.6	80.5	84.4	90.7	95.0	96.1	97.5	93.8	88.2	79.7	74.9
		MCWB	64.7	65.9	67.2	69.8	73.4	76.0	77.4	77.1	74.9	72.7	68.3	67.8
	5%	DB	70.3	73.0	77.2	82.1	88.8	92.5	93.7	95.2	91.4	84.9	75.9	72.0
		MCWB	63.6	64.8	65.7	68.8	73.1	76.2	77.8	77.2	74.9	70.5	66.0	66.4
	10%	DB	66.4	70.1	74.1	79.9	86.7	90.5	91.9	92.9	89.7	82.0	72.8	68.2
		MCWB	61.0	63.8	64.6	67.7	72.8	76.1	77.7	77.1	74.6	69.6	65.3	63.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.1	70.3	72.0	75.3	79.0	80.4	81.5	81.9	80.2	77.9	73.5	71.9
		MCDB	73.9	75.6	77.2	81.8	87.1	89.3	90.1	90.4	87.6	85.0	78.6	75.5
	2%	WB	67.1	68.7	70.0	73.3	77.0	79.2	80.3	80.3	78.5	76.1	71.2	69.5
		MCDB	71.3	74.2	76.0	79.7	86.0	88.6	89.8	89.8	87.2	83.2	75.7	74.0
	5%	WB	65.2	66.8	68.5	72.0	75.6	78.3	79.4	79.4	77.5	74.7	69.1	67.5
		MCDB	68.9	71.9	74.4	78.8	84.0	87.7	89.4	89.4	86.3	81.3	74.4	71.3
	10%	WB	62.5	64.6	66.8	70.5	74.5	77.6	78.6	78.6	76.4	72.8	66.5	64.2
		MCDB	65.8	68.9	72.4	77.0	82.6	86.9	88.2	88.3	84.7	78.7	71.6	67.3
Mean Daily Temperature Range	5% DB	MDBR	19.8	19.9	20.8	21.3	20.3	18.9	18.5	19.9	21.0	23.6	22.5	19.7
		MCDBR	22.0	21.5	21.8	22.1	21.3	21.5	20.8	23.2	23.0	24.0	24.6	21.7
		MCWBR	15.0	12.6	11.3	10.2	7.4	6.6	6.5	6.6	7.7	10.3	14.3	15.1
	5% WB	MCDBR	17.5	18.2	18.7	17.9	18.0	18.2	18.6	19.5	18.8	18.5	20.4	19.3
		MCWBR	14.0	12.7	11.8	10.2	7.9	7.3	7.0	7.0	7.3	9.5	14.1	14.7
Clear-Sky Solar Irradiance	taub		0.331	0.352	0.372	0.413	0.440	0.479	0.500	0.494	0.443	0.372	0.355	0.343
	taud		2.522	2.467	2.407	2.295	2.265	2.196	2.166	2.182	2.296	2.467	2.469	2.488
	Ebn at Noon		285	288	289	279	271	259	253	253	263	277	272	273
	Edh at Noon		28	32	36	42	43	47	48	46	40	31	29	27
All-Sky Solar Radiation	RadAvg		875	1000	1390	1717	1907	1922	1911	1801	1589	1351	1009	774
	RadStd		80	173	114	134	128	159	147	166	146	187	102	88

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+212	+139

Nomenclature: See separate page